

Course Orientation & Data Science Process

What data science is and how a project runs



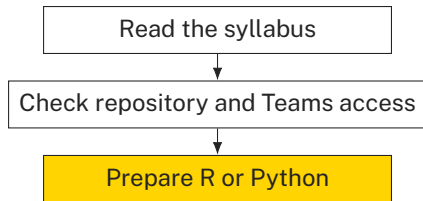
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Faculty of Computer Science
spadon@dal.ca

Three Things to Do This Week

- Read the **syllabus**, including assessment and integrity policies.
- Check your access to the **course repository and Teams**.
- Prepare one working **R or Python environment** for Thursday.

Course materials are public.
Use Teams for
course questions.



Submit the ungraded syllabus and integrity acknowledgement by September 22.

How and Where We Meet

- Tue and Thu, **16:05–17:25**.
- Carleton Tupper Building, #**3H01**.
- Term runs for **twelve teaching weeks**; see the official calendar.
- We do not have lab sessions. Labs are integrated into the classes.

Shared Expectations

- Complete your work on time.
 - Respect other people's space.
 - Bring your questions to class.
 - Communicate respectfully.
-

Today's Data Science (DS) Questions

Why Data Science?

1

Where the demand comes from, and what the work in a data-related job is.

What Is Data Science?

2

The shapes data arrives in, one clinical table, and a definition to hold them.

What Is the Process Like?

3

CRISP-DM, the six-stage loop a real project runs, and its questions.

How Does This Course Work?

4

Labs, assignments, examinations, and where every course document is posted.

Why Learn Data Science?

Demand, data, and computing power have grown together, and Data Science sits between them.



A 2022 Job-Market Snapshot

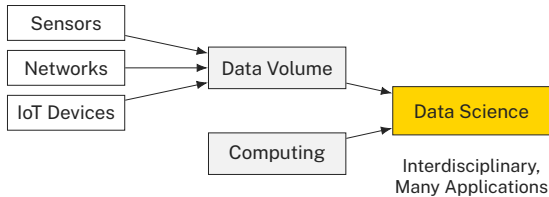
Demand for data-related roles is high across the whole ranking, not only at the top.

Rank	Job	Average Salary	Growth (%) in Postings, 2019–22
1	Data engineer	\$85,706	242
2	Site reliability engineer	\$90,010	218
3	Engineering manager	\$111,515	161
... ranks 4 to 18, fifteen further roles ...			
19	Data scientist	\$76,938	60

Indeed's twenty best jobs in Canada, 2022, reported by Canadian HR Reporter. hrreporter.com

What Is Driving the Field

- Increasing volumes of data
 - Variety of cheap sensors
 - Increased networking
 - Internet of Things
- Increasing computing power
- An interdisciplinary field
- A variety of applications



Application Domains

Healthcare



Diagnosis support, risk scoring, and planning.

Finance



Credit decisions, pricing, and fraud.

Smart Cities



Mobility, demand, and infrastructure planning.

E-Commerce



Recommendation and demand forecasting.

Social Networks



Communities, influence, and information flow.

Environment



Monitoring, forecasting, and mitigation.

Data Science by Example

Data of every kind, one table of rows and columns, and a definition of the field.



Increasing Volumes of Different Types

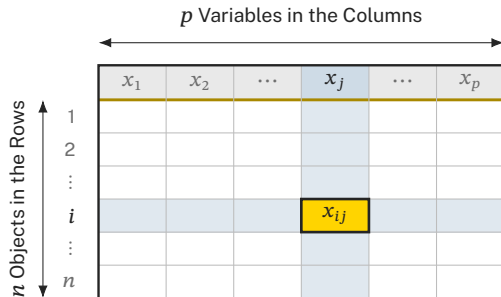
- Tabular Data
- Databases of Related Tables
- Text
- Multimedia: Images, Video, Sound
- Data Streams and Time Series

In This Course:
Tabular Data

The other types matter, and the reasoning here transfers to them.

Tabular Data

- A data set can be represented as an $n \times p$ matrix with n observations or cases, each described by p variables.
- Rows represent the cases, also called objects, examples, instances, or records.
- Columns represent variables, also called features, fields, or attributes.
- The entry x_{ij} is the value of variable j for case i . A table entry may be numeric or categorical.



The Heart Disease Data Set

Row	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
	numeric	category	category	numeric	numeric	category
1	40	M	ATA	140	289	0
2	49	F	NAP	160	180	0
3	37	M	ATA	130	283	0
4	48	F	ASY	138	214	0
5	54	M	NAP	150	195	0
6	39	M	NAP	120	339	0

12 attributes and **918 cases**. The first six patients, with the first six variables. Types describe their analytical roles.

UCI Heart Disease data, distributed on [kaggle.com/datasets/fedesoriano/heart-failure-prediction](https://www.kaggle.com/fedesoriano/heart-failure-prediction)

The Same Patients, More Variables

Row	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
	category	numeric	category	numeric	category	category
1	Normal	172	N	0.0	Up	0
2	Normal	156	N	1.0	Flat	1
3	ST	98	N	0.0	Up	0
4	Normal	108	Y	1.5	Flat	1
5	Normal	122	N	0.0	Up	0
6	Normal	170	N	0.0	Up	0

Row numbers match the preceding slide. HeartDisease is the categorical target, encoded as 0 or 1. Numeric storage does not make a category continuous.

Reading the Same Table Twice

R

```
heart <- read.csv(  
  "heart.csv",  
  stringsAsFactors = TRUE  
)  
dim(heart)      # 918 12  
head(heart)  
str(heart)
```

Python

```
import pandas as pd  
  
heart = pd.read_csv("heart.csv")  
heart.shape      # (918, 12)  
heart.head()  
heart.dtypes
```

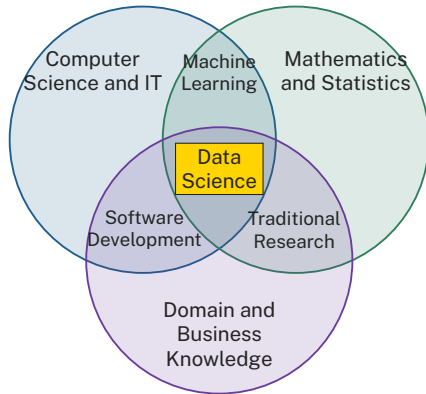
Both calls produce 918 rows and 12 columns. R imports text as factors here. Pandas text dtypes depend on its version. Integer-coded categories need explicit conversion before modelling.

What Is Data Science?

“Data mining is the analysis of (often large) observational data sets to find unsuspected relationships and to summarize the data in novel ways that are both understandable and useful to the data owner.”

Hand, Mannila, and Smyth, *Principles of Data Mining*, MIT Press, 2001, page 1.

Observational means the researcher does not assign an intervention. Such data may be collected for the research question itself. Data science also uses experiments.

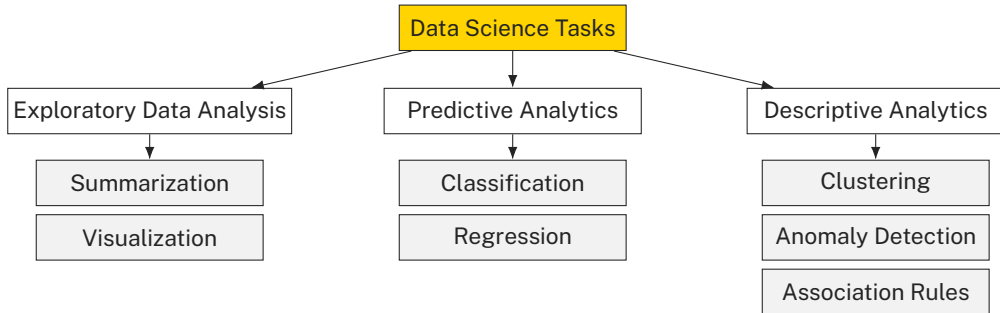


Data Science Tasks

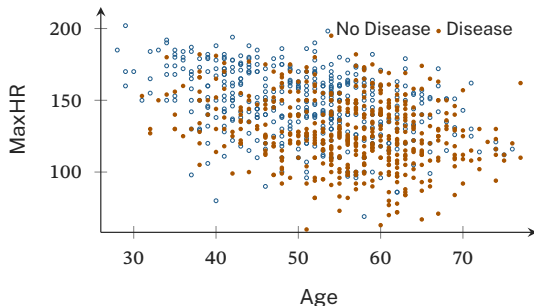
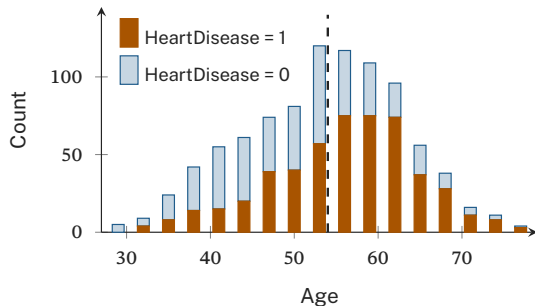
Every question you can ask of a table falls into one of a small number of task families.



Data Science Tasks

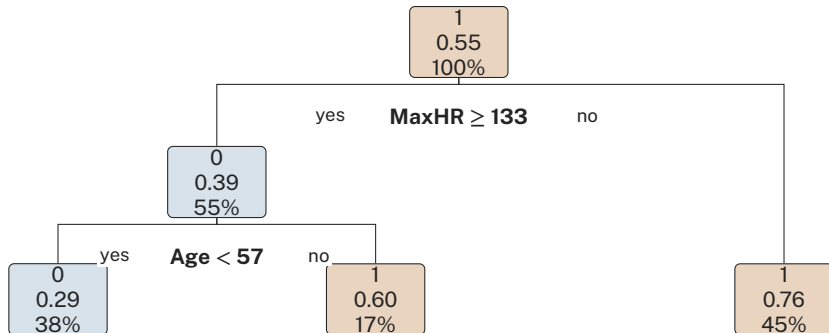


Exploratory Data Analysis



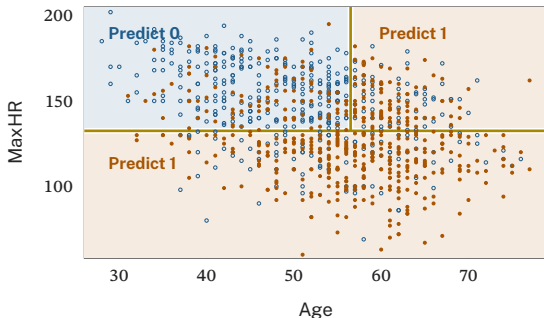
All 918 patients. Three-year age bins start at 27.5; the dashed median is 54. The scatter uses unjittered observations, so coincident points overlap. Neither plot fits a model.

Predictive Analytics, the Fitted Tree



Each node reports the predicted class, the share of disease cases inside it, and the share of training data it holds. The root predicts 1 because 55% of the 918 patients are ill.

The Partition the Tree Induces



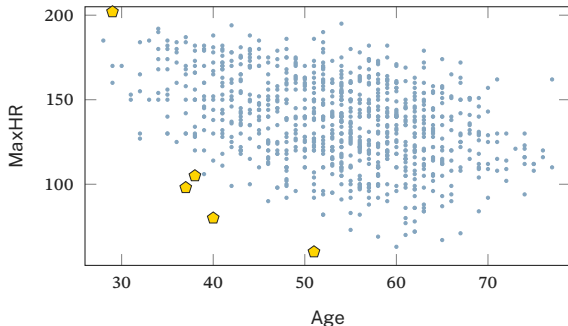
- Two splits cut the plane in three.
- Low maximum heart rate predicts disease on its own.
- Above the heart-rate threshold, age decides.

Training accuracy: 71.24%

654 of 918 training cases. Split boundaries are 132.5 and 56.5, equivalent to the integer rules shown. Future performance is covered in Week 9.

Descriptive Analytics, an Example

Five Most Distant Cases, K-Means



- No target variable is used.
- K-means groups the patients, then distance to the nearest centre ranks them.
- CSV rows 830, 3, 508, 391, and 569 are flagged, in decreasing distance order.

Fixed initial centres and population-SD scaling make this example reproducible. Flagged means unusual, not necessarily wrong.

Classification in R and Python

R

```
hist(heart$Age) # EDA
fit <- rpart::rpart(
  factor(HeartDisease) ~ Age + MaxHR,
  data = heart, method = "class",
  control = rpart::rpart.control(
    maxdepth = 2, cp = 0.01, xval = 0)
)
mean(predict(fit, type = "class") ==
  heart$HeartDisease)
```

Python

```
from sklearn.tree import (
    DecisionTreeClassifier
)
heart["Age"].hist() # EDA
X = heart[["Age", "MaxHR"]]
y = heart["HeartDisease"]
fit = DecisionTreeClassifier(
    max_leaf_nodes=3,
    random_state=3141
).fit(X, y)
fit.score(X, y)
```

Continue from the import example. R's depth and pruning controls and Python's three-leaf limit yield the displayed rules on this dataset. Both score 654/918; they need not agree on other data.

K-Means in R and Python

R

```
X <- as.matrix(heart[c("Age", "MaxHR")])
Z <- scale(X, scale = FALSE)
Z <- sweep(Z, 2, sqrt(colMeans(Z^2)), "/")
km <- kmeans(
  Z, centers = Z[c(1, 301, 601), ],
  algorithm = "Lloyd", iter.max = 300
)
d <- sqrt(rowSums(
  (Z - km$centers[km$cluster, ])^2
))
order(-d, seq_along(d))[1:5]
```

Population-SD scaling, fixed initial rows, and Lloyd updates. Both rank CSV rows 830, 3, 508, 391, 569 first. Row numbers exclude the header.

Python

```
import numpy as np
from sklearn.cluster import KMeans
X = heart[["Age", "MaxHR"]].to_numpy()
Z = (X - X.mean(0)) / X.std(0, ddof=0)
km = KMeans(
    n_clusters=3, init=Z[[0, 300, 600]],
    n_init=1, algorithm="lloyd",
    max_iter=300, tol=0).fit(Z)
centres = km.cluster_centers_[km.labels_]
d = np.linalg.norm(Z - centres, axis=1)
np.argsort(-d, kind="stable")[:5] + 1
```

CRISP-DM, the Typical Data Science Process

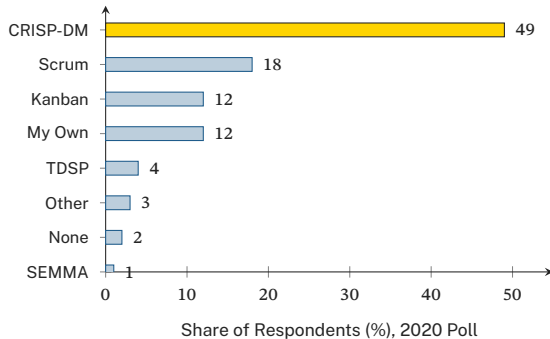
A project is a loop with six named stages, not a straight line from data to model.



CRISP-DM

- **C**Ross Industry **S**tandard
Process for **D**ata **M**ining.
- The most selected option in this practitioner poll.

The 2020 convenience poll has 109 respondents. It does not estimate prevalence among all practitioners. Rounded percentages sum to 101%.



Data Science Process Alliance poll, 2020, 109 respondents. datascience-pm.com

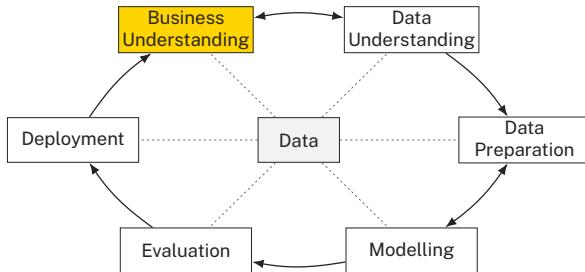
Business Understanding

1. Business Understanding

What are the business goals?

How can data science help achieve them?

Nothing later in the loop can repair a project that never named the decision it supports.

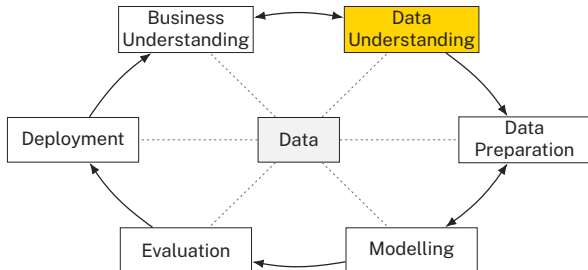


Data Understanding

2. Data Understanding

What data do we have or need?
Are there data quality issues?

Can you think of any potential data quality issues or limitations?

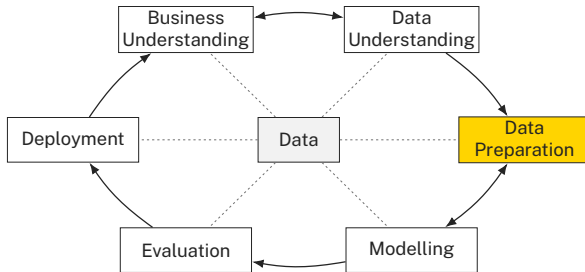


Data Preparation

3. Data Preparation

How do we clean and transform data for modelling?

In practice this stage consumes most of the project's time, and every choice made here is an assumption the model inherits.

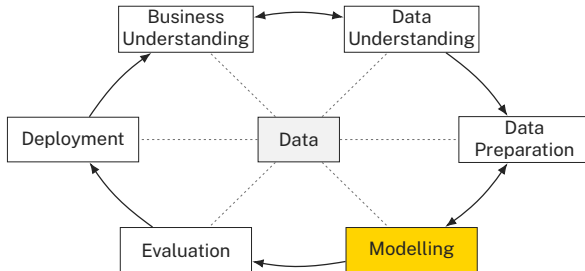


4. Modelling

What modelling techniques should we use?

How do we calibrate them?

The two-headed arrow back to preparation is not decoration. Model choice changes what the data must look like.

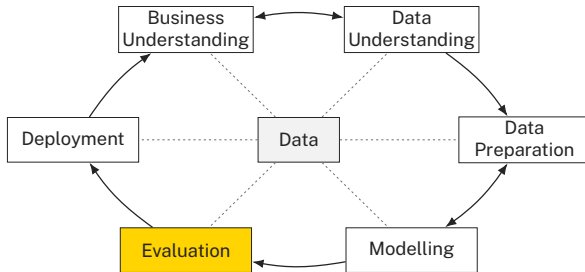


Evaluation

5. Evaluation

Does the selected solution meet the business goals?

This is a business question, not only an accuracy number. A model can score well and still fail the goal set in stage one.

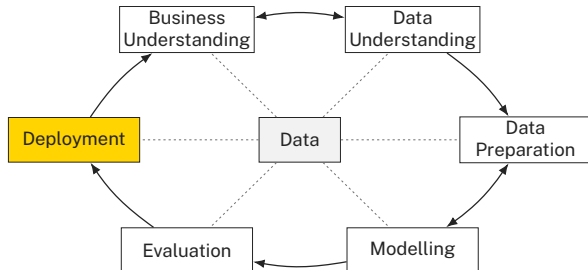


Deployment

6. Deployment

How do we deliver our solution to the business?

What kind of deliverables would provide value to stakeholders?



Think, Pair, Share

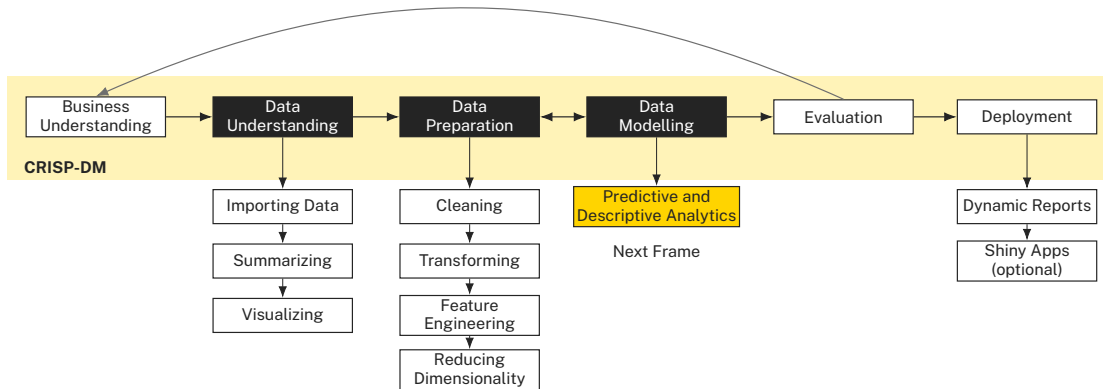
1. Take a minute to think of a data science application you are curious about.
 - What data would you need? Do you think it would be readily available?
 - What challenges do you expect to face?
2. Discuss with your neighbour.
3. Volunteer to share with the class.

About the Course

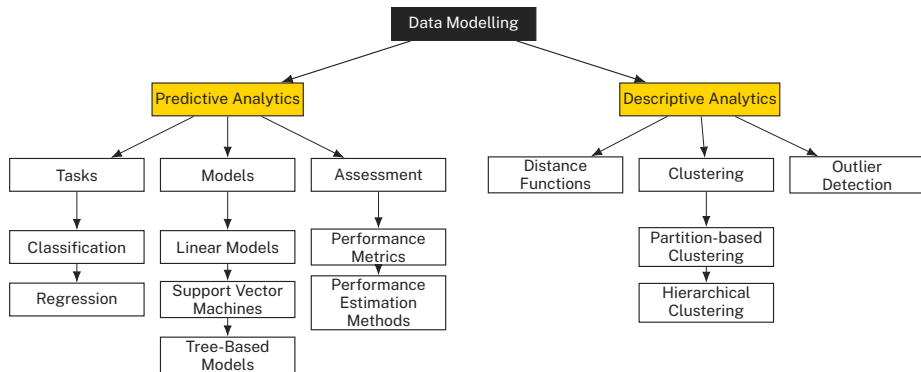
The syllabus follows the same loop you have just seen, one stage at a time.



The Process Spine and the Data Stages



The Modelling Branch



What You Will Be Able to Do

1. Describe the main steps in a data science project.
2. Transform data to prepare it for analysis and modelling.
3. Explore data using appropriate statistics and visualizations.
4. Design and implement a data science application, using predictive and descriptive modelling techniques.
5. Evaluate, compare, and select models for a dataset.
6. Prepare reproducible reports in R Markdown or Python notebooks.

A Working Environment for Thursday

The R Toolchain

- Use **Posit Cloud** or an R notebook on Kaggle.
- For local work, install **R** from CRAN and an editor such as **RStudio**.
- Packages: tidyverse, rpart, cluster.

Bring one working local or cloud environment. Both languages use the same rubric. Contact the instructor early if computer access is a barrier.

The Python Toolchain

- Use **Google Colab** or a Python notebook on Kaggle.
- For local work, install **Python 3**, create a virtual environment, and choose an editor.
- Use pandas and seaborn for data and plots; scikit-learn for models.

cran.r-project.org posit.co/download/rstudio-desktop python.org/downloads code.visualstudio.com

Choose One Free Notebook Environment

Service	Language	Environment
Posit Cloud	R or Python	RStudio or Jupyter projects
Kaggle Notebooks	R or Python	Browser notebooks
Google Colab	Python	Browser notebooks

- Local R or Python is equally acceptable. No paid service is required.
- Free services have usage limits. Save your files and record dependencies.
- Keep assessed notebooks private until the deadline and release to the class.

posit.cloud kaggle.com/code colab.research.google.com

How We Stay in Contact

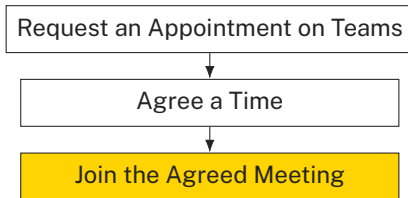
- Announcements go in the repository.
 - Activate e-mail notifications.
 - Check your Dalhousie e-mail daily.
- Use Teams for class questions and to arrange office hours.

For Private Matters, Use E-mail

- Subject line: [CSCI 3141] Subject matter.
 - Instructor: Dr. Gabriel Spadon, spadon@dal.ca.
 - Replies normally take three business days.
-

By Appointment

- Office hours are held **by appointment**.
- Arrange the appointment through **Microsoft Teams**.
- Write ahead with the subject you would like to discuss.



Format

Arrange through Teams



Arrange a time on Teams.

Individually or in Small Groups



For shared questions, request a group appointment.

Short and Focused



Say what you want to cover when you write, so the meeting starts at the question.

Early Beats Late



A five-minute conversation in week two saves an hour in week ten.

Academic Integrity and Conduct

- Academic integrity is expected from all students at Dalhousie.
- Review the Academic Integrity resources in the course repository.
- Originality checking may be used. Notify the instructor by September 22 to use the syllabus's equivalent alternative without penalty.
- Everyone at Dalhousie should be treated with dignity and respect.
- Keep this in mind in every interaction, in class and online.

Generative AI is prohibited on integrated labs and examinations.
Assignment use requires express authorization in its instructions.

Dalhousie academic integrity, dal.ca/about/leadership-governance/academic-integrity.html

How Class Time Is Used

- Classes run **16:05 to 17:25** on Tuesdays and Thursdays in Carleton Tupper 3H01.
- Attendance is recommended.
- Course materials are published in the course GitHub repository.

Active Learning

- In-class response questions using the polling tool linked from the course repository.
- Work in pairs or small groups.
- Thursday classes start the integrated weekly labs.

Integrated Labs (15%)

- Each lab applies the week's method and carries earlier decisions forward.
- There is no separate lab section. The labs start inside the Thursday classes.
- One rubric covers dataset, analysis, reading, and reproducibility.
- The **best 10 of 12** lab results count, so two can be missed without penalty.
- Submit by **23:59 two calendar days** after the lab begins.

1	2	3	4	5	6
7	8	9	10	11	12

Twelve Offered, Best Ten Counted

No quizzes are scheduled or included in the grading scheme.

Assignments (45%)

Assignment	Weight	What It Asks For	Due
Assignment 1	7.5%	Import data and perform basic data exploration	11 October 2026
Assignment 2	7.5%	Pre-process and explore data	1 November 2026
Assignment 3	15%	Design and implement a predictive modelling solution	29 November 2026
Assignment 4	15%	Design and implement a descriptive modelling solution	9 December 2026

The four assignments trace the syllabus arc, from import to prepare to predict to describe.

More on Assignments

- Assignments are **individual** work.
- They are due at 23:59 on the announced date.
- Submit the full commit SHA and its URL privately. Keep work private until the deadline and the instructor opens sharing.

Student Declaration of Absence and accommodation provisions apply as the University regulations set them out.

Late Submissions

- Assignments 1 to 3 lose **5 percentage points per late calendar day**, up to seven days.
- Later work requires an approved absence or accommodation process.
- Assignment 4 has no routine late window.

From Private Work to Public Portfolio

1. Work privately

Use a private repository with instructor access.
Commit regularly.
Keep your work private.

2. Submit by the deadline

Send the full commit SHA and commit URL through the announced private channel. The receipt and saved snapshot fix the assessed version.

3. Share after release

After the deadline and when the instructor opens sharing, publish that same commit and its history.
Share it in class.

A branch in a public repository is public too. Use the syllabus's private-portfolio alternative when required.

Keep the Evidence and Credit the Source

Preserve assessment history

- Keep the submitted commit reachable permanently.
- No rewriting, squashing, deletion, force pushes, or backdating.
- Make corrections in new commits. Git dates alone do not prove timely submission.

Share released materials

- Credit the author and CSCI 3141. Preserve source links and licence terms.
- Keep instructor-marked private material, grades, and feedback private.
- Do not share assessed work early or share another student's work.

Midterm and Final Examination (40%)

Midterm (15%)

- Held **in class**, in the week the calendar names.
- Covers everything up to that class.

Final Examination (25%)

- Scheduled by the Registrar.
- **Cumulative**, covering everything.

Bring valid Dalhousie photo ID. Course and personal notes are allowed unless the examination lists narrower approved aids. Internet access, communication, and generative AI are prohibited.

Grading Summary

Component	Weight	Rule
Integrated labs	15%	best 10 of 12 results
Assignment 1	7.5%	individual
Assignment 2	7.5%	individual
Assignment 3	15%	individual
Assignment 4	15%	individual
Midterm	15%	in class, 20 October 2026
Final examination	25%	cumulative, Registrar-scheduled
Total	100%	

The weighted total, the combined assignment component, and the final examination must each reach 50%.

Dates to Put in Your Calendar Now

Course Dates

Event	Date
Classes begin	8 Sep
Assignment 1	11 Oct
Midterm, in class	20 Oct
Assignment 2	1 Nov
Assignment 3	29 Nov
Last class	3 Dec
Assignment 4	9 Dec

University Deadlines

Deadline	Date
Last day to add	22 Sep
Drop without a W	7 Oct
Drop with a W	5 Nov
Fall Break	9–13 Nov
Final examination	set by the Registrar

The course GitHub repository carries the authoritative version if anything changes.

University dates, dal.ca/study/plan-your-degree/important-dates.html

Course Texts and Resources

- There is **no required paid textbook**.
The repository slides, notes, and lab handouts are the primary source.
- The texts listed here are optional.
Check library access before obtaining a copy.
- Cheatsheets and documentation links are posted with each module.

Optional Further Reading

Rigdon, S. E., Fricker, R. D., and Montgomery, D. C., *Introduction to Probability and Statistics for Data Science with R*, Cambridge University Press, 2024.

Shah, C., *A Hands-On Introduction to Data Science with Python*, 2nd edition, Cambridge University Press, 2026.

Torgo, L., *Data Mining with R: Learning with Case Studies*, 2nd edition, CRC Press, 2017.
Earlier editions are acceptable.

doi.org/10.1017/9781316286166 doi.org/10.1017/9781009588911 libraries.dal.ca

What Today's Class Covered

Why Data Science

1

Demand runs the length of the postings ranking, and most of the work is making data fit to use.

What Data Science Is

2

Learning from data through computing, statistics, and domain knowledge. This course focuses on tables.

Three Kinds of Question

3

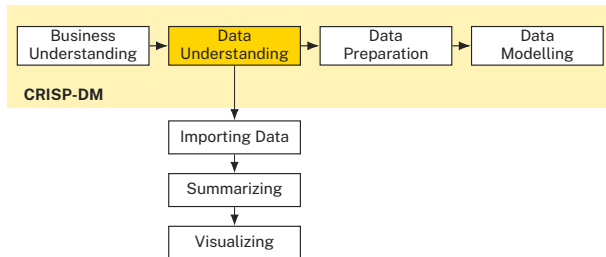
Exploratory, predictive, and descriptive, asked of one Heart Disease table inside the CRISP-DM loop.

How This Course Works

4

Twelve integrated labs, four assignments, a midterm, and a cumulative final.

Data Understanding



We start at the second stage, importing, summarizing, and visualizing a data set.

Read the syllabus. Bring R or Python ready to use.